

Wardle Editorial Canon

Standards for Governed Editorial Artifacts

Part V — Knowledge and Reasoning Governance

1. Purpose

Knowledge and Reasoning Governance defines how Wardle represents, evaluates, connects, and maintains the clinical knowledge underlying diagnostic education.

Cases and Diagnosis Education communicate knowledge to learners.

The Knowledge and Reasoning layer represents the clinical relationships that make those educational artifacts coherent.

It enables WEOS to govern:

- findings associated with diagnoses
- discriminating evidence
- important mimics
- diagnostic contrasts
- complication signals
- management signals
- escalation triggers
- reasoning pathways
- differential relationships
- extracted knowledge candidates
- unsupported claims
- knowledge provenance

Its purpose is not to build a graph for technical sophistication.

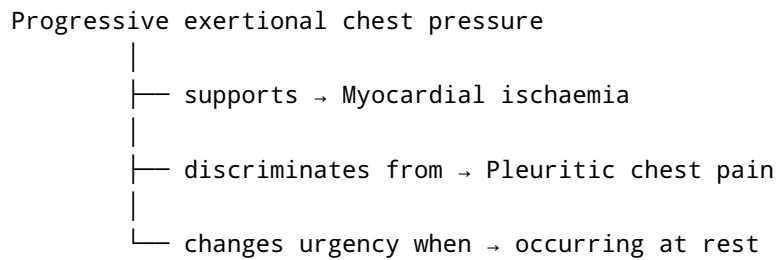
Its purpose is to make clinical reasoning explicit, reusable, reviewable, and governable.

2. Knowledge Governance Philosophy

Clinical knowledge should not remain trapped inside isolated paragraphs, case explanations, or generation prompts.

Where clinically meaningful relationships can be represented explicitly, WEOS should preserve them as governed knowledge.

For example:



This representation allows the same knowledge to support:

- case generation
- case review
- Diagnosis Education
- differential analysis
- recall prompts
- quality assessment
- curriculum coverage
- maintenance

Explicit representation does not replace clinical narrative.

It provides the governed reasoning structure beneath it.

3. Foundational Distinctions

Knowledge Governance depends upon several distinctions.

3.1 Extracted statement and governed fact

An extracted statement is knowledge identified within an existing artifact.

A governed fact is knowledge that has completed the required editorial decision process.

Extraction establishes provenance.

It does not establish authority.

3.2 Association and reasoning relationship

An association indicates that two concepts are connected.

A reasoning relationship explains how the connection affects clinical interpretation.

For example:

Association:

Hyperpigmentation is associated with Addison disease.

Reasoning relationship:

Diffuse hyperpigmentation, particularly involving palmar creases and buccal mucosa, supports primary adrenal insufficiency over secondary adrenal insufficiency because elevated adrenocorticotrophic hormone is expected in primary disease.

The second representation provides greater educational value because it explains discrimination.

3.3 Graph fact and teaching relationship

A Graph Fact represents a governed diagnosis-level knowledge statement.

A Teaching Relationship represents how two diagnoses should be compared or connected for educational purposes.

A graph fact may state:

Haemoptysis may occur in chronic pulmonary aspergillosis.

A teaching relationship may state:

Chronic pulmonary aspergillosis is commonly confused with pulmonary tuberculosis in patients with chronic constitutional and respiratory symptoms; pre-existing cavities with fungal evidence support aspergillosis.

These artifacts may support one another but should not be treated as identical.

3.4 Evidence and evidence source

An Evidence Node represents a clinical finding or concept.

An external reference supports the scientific authority of a claim.

A case may demonstrate an evidence relationship.

A case does not independently establish clinical evidence authority.

WEOS must therefore distinguish:

- clinical evidence concept
- diagnosis–evidence relationship
- supporting Wardle artifact
- external supporting source

3.5 Differential text and governed differential relationship

A differential name written in a case or education section is authored content.

A differential mapping resolves that text to a Diagnosis Registry identity.

A differential link records a governed relationship between the source artifact and the resolved diagnosis.

A teaching relationship may then explain why the diagnoses are confused and how they should be distinguished.

Each layer serves a different function.

4. Knowledge Artifact Families

Knowledge and Reasoning Governance includes:

- Graph Candidates
- Graph Facts
- Evidence Nodes
- Diagnosis–Evidence Relationships
- Teaching Relationships
- Differential Mappings
- Differential Links
- Reasoning Paths
- Diagnostic Comparisons
- Reasoning Draft Validation
- Unsupported-Claim Signals
- Knowledge-Governance Decisions

These artifacts collectively form the governed reasoning foundation of WEOS.

5. Graph Candidate

5.1 Purpose

A Graph Candidate is a proposed unit of structured knowledge extracted from or suggested by an existing artifact.

Graph Candidates allow Wardle to identify potentially reusable knowledge without automatically treating the source text as institutionally trusted fact.

5.2 Sources

A Graph Candidate may originate from:

- a Clinical Case
- a Case Explanation
- Diagnosis Education
- a generated educational section
- an editor suggestion
- a future imported source

The candidate should preserve the exact artifact and version from which it originated.

5.3 Candidate types

Current canonical Graph Candidate types include:

- Finding
- Investigation
- Mimic
- Pitfall
- Management
- Complication
- Recall Prompt
- Case Reasoning

Each type should have an artifact-specific quality standard.

5.4 Finding candidate

A Finding Candidate represents a clinically meaningful symptom, sign, risk factor, historical feature, or observed pattern.

It should identify:

- the finding
 - its diagnosis context
 - its clinical significance
 - its source
 - any target diagnosis where relevant
-

5.5 Investigation candidate

An Investigation Candidate represents a diagnostic test, result, interpretation, or testing principle.

It should distinguish between:

- test name
 - expected finding
 - diagnostic role
 - limitation
 - discriminator value
-

5.6 Mimic candidate

A Mimic Candidate proposes that another diagnosis is educationally relevant as a source of confusion or differential consideration.

A mimic candidate should not be approved solely because the alternative diagnosis appears in a differential list.

Review should determine:

- whether confusion is clinically plausible
 - which presentation is shared
 - which discriminator is educationally important
 - whether the relationship is directional or reciprocal
-

5.7 Pitfall candidate

A Pitfall Candidate represents a possible learner or clinical reasoning error.

It should state:

- the mistaken assumption or action
 - why the error is plausible
 - the consequence
 - the safer reasoning principle
-

5.8 Management candidate

A Management Candidate represents a diagnosis-level management principle.

It should not be promoted without sufficient context regarding:

- indication
 - timing
 - severity
 - contraindication
 - escalation
 - patient population
-

5.9 Complication candidate

A Complication Candidate represents an important adverse consequence of the diagnosis or its delayed treatment.

It should distinguish:

- direct disease complication
 - treatment complication
 - associated condition
 - severity marker
-

5.10 Recall Prompt candidate

A Recall Prompt Candidate represents potentially reusable active-recall content.

Promotion into governed recall content should require compliance with the Recall Prompt standard in Part IV.

5.11 Case Reasoning candidate

A Case Reasoning Candidate represents a reasoning insight derived from case clue progression or explanation.

It may capture:

- a discriminator
- differential elimination
- escalation recognition
- clue-order principle
- diagnostic turning point

Case-specific reasoning should not automatically be generalised to all presentations of the diagnosis.

6. Graph Candidate Provenance

Every Graph Candidate should preserve:

- diagnosis
- candidate type
- source artifact type
- source artifact identity
- source version
- source path
- raw source text
- normalised text
- structured payload where relevant
- target diagnosis where relevant
- unresolved target text where necessary
- extraction confidence
- reviewer
- review date
- review note
- merge destination where applicable
- promoted Graph Fact where applicable

Provenance should make it possible to trace every governed Graph Fact back to the content from which it emerged.

7. Graph Candidate Normalisation

Graph Candidates should undergo controlled normalisation to support:

- duplicate detection
- clustering
- consistent labels
- target diagnosis resolution
- merge decisions

Normalisation may address:

- letter case
- punctuation
- superficial phrasing differences
- accepted terminology variation

Normalisation must not remove clinically meaningful qualifiers.

Examples include:

- acute versus chronic
- unilateral versus bilateral
- painless versus painful
- proximal versus distal
- exertional versus resting
- early versus late

These qualifiers may determine the clinical meaning of the candidate.

8. Graph Candidate Deduplication

Candidates may be duplicates where they express the same clinical knowledge from different sources.

Duplicate analysis should consider:

- diagnosis
- candidate type
- normalised meaning
- target diagnosis
- clinical qualifiers
- source context
- structured payload

Candidates should not be merged merely because they share keywords.

For example:

- “Chest pain” and “pleuritic chest pain” are not equivalent.
- “Normal troponin” and “serially normal troponin” may carry different diagnostic meaning.
- “Weakness” and “progressive proximal muscle weakness” should not be collapsed.

9. Graph Candidate Lifecycle

The canonical Graph Candidate lifecycle is:

Candidate

↓

Editorial Review

↓

Approved, Rejected, or Merged

↓

Promoted to Graph Fact where appropriate

9.1 Candidate

The proposed knowledge exists but remains ungoverned.

9.2 Approved

The candidate has been accepted as valid structured knowledge.

Approval should identify whether promotion into a Graph Fact is immediate or separately controlled.

9.3 Rejected

The candidate has not been accepted.

Rejection may occur because it is:

- clinically inaccurate
- too generic

- duplicative
- improperly scoped
- unsupported
- not educationally useful
- incorrectly classified

9.4 Merged

The candidate has been consolidated into another candidate or existing fact.

The original candidate and source provenance should remain traceable.

10. Graph Fact

10.1 Purpose

A Graph Fact is a governed diagnosis-level knowledge statement available for use across WEOS.

It represents accepted structured knowledge rather than unreviewed extraction.

10.2 Required structure

A Graph Fact should preserve:

- diagnosis
 - fact type
 - label
 - normalised label
 - structured payload
 - target diagnosis where relevant
 - operational status
 - source candidate
 - provenance
 - creation and maintenance history
-

10.3 Active status

An Active Graph Fact may influence:

- editorial workspace projections
- generation context

- case validation
- education assessment
- reasoning paths
- teaching relationships
- recall generation
- coverage analysis

Active status must follow governance.

It should not result merely from successful extraction.

10.4 Archived status

A Graph Fact may be archived where it is:

- outdated
- superseded
- duplicated
- no longer educationally relevant
- inconsistent with current evidence
- based on a deprecated diagnosis relationship

Archival should preserve provenance and prior use.

10.5 Fact scope

Every Graph Fact should clearly identify its scope.

Possible scope qualifiers include:

- typical
- early
- late
- severe
- uncommon
- paediatric
- adult
- pregnancy-associated
- complication-specific
- setting-specific

A fact that is true only in a subgroup must not be represented as universally applicable.

11. Evidence Node

11.1 Purpose

An Evidence Node represents a reusable clinical concept that may participate in reasoning across multiple diagnoses.

Examples include:

- exertional chest pressure
- hyperreflexia
- elevated serum lipase
- cavitary lung lesion
- recent antibiotic exposure
- haemodynamic instability
- painless haematuria

The Evidence Node defines the finding.

It does not determine what the finding means for every diagnosis.

Meaning is established through Diagnosis–Evidence Relationships.

11.2 Evidence types

Canonical Evidence Types include:

- Symptom
- Examination
- Laboratory
- Imaging
- Risk Factor
- History
- Management
- Complication
- Epidemiology

Evidence type should describe the nature of the concept rather than how it happened to be presented in one case.

11.3 Clinical category

Evidence Nodes may also belong to broad clinical categories such as:

- pain
- bleeding
- infection
- neurological
- respiratory
- cardiovascular
- gastrointestinal
- endocrine
- renal
- trauma
- other

Clinical categories support organisation and reasoning analysis.

They should not replace evidence type.

11.4 Node identity

Every Evidence Node should possess:

- stable identity
- normalised key
- display label
- evidence type
- clinical category
- accepted synonyms
- governance status

The display label should remain clinically meaningful outside one source sentence.

11.5 Evidence Node lifecycle

An Evidence Node may be:

- Candidate
- Active
- Rejected
- Deprecated

An Active Evidence Node is available for governed relationships.

Activation does not imply that the node supports every diagnosis to which it has been tentatively connected.

Each relationship requires independent governance.

12. Diagnosis–Evidence Relationship

12.1 Purpose

A Diagnosis–Evidence Relationship defines how an Evidence Node affects reasoning about a diagnosis.

It transforms a clinical finding into explicit diagnostic meaning.

12.2 Relationship types

Current canonical relationship types include:

- Supports
 - Discriminates
 - Escalates
 - Rules Out
 - Complication Signal
 - Management Signal
-

12.3 Supports

A supporting relationship indicates that the evidence increases the plausibility of the diagnosis.

Support should be appropriately qualified.

A common non-specific symptom may provide weak support.

A characteristic pattern may provide strong support.

Support does not necessarily establish diagnosis.

12.4 Discriminates

A discriminating relationship indicates that the evidence helps distinguish the diagnosis from one or more plausible alternatives.

A discriminator should identify:

- the target diagnosis
 - relevant competing diagnosis or group
 - direction of discrimination
 - clinical reasoning summary
 - strength
-

12.5 Escalates

An escalating relationship indicates that the evidence changes urgency, level of care, monitoring, or intervention.

Examples include:

- haemodynamic instability
- altered mental state
- respiratory failure
- peritonism
- rapidly progressive neurological deficit

Escalation evidence may support management even when it does not strongly identify the diagnosis.

12.6 Rules Out

A rules-out relationship indicates that evidence materially weakens or excludes the diagnosis.

This relationship should be used cautiously.

Most clinical findings do not absolutely rule out a diagnosis.

Where exclusion is incomplete, the relationship should reflect reduced probability rather than certainty.

12.7 Complication Signal

A complication-signal relationship indicates that the evidence suggests a recognised complication of the diagnosis.

The relationship should distinguish between:

- presence of the primary diagnosis
 - severity of the diagnosis
 - presence of a complication
-

12.8 Management Signal

A management-signal relationship indicates that the evidence changes treatment, referral, monitoring, or intervention.

It should preserve the clinical context in which the management implication applies.

13. Relationship Strength

Diagnosis–Evidence Relationships may carry strength.

Strength should reflect the clinical and educational weight of the relationship.

It may consider:

- sensitivity
- specificity
- prevalence
- clinical context
- discriminator value
- severity relevance
- evidence quality

Strength must not create false mathematical precision where evidence is qualitative.

14. Discriminator Weight

Discriminator weight represents the relative value of the evidence in distinguishing the diagnosis from important alternatives.

A finding may strongly support several diagnoses while possessing low discriminator value.

For example:

Fever may support infection but may not distinguish between several infectious diagnoses.

A narrower finding may possess greater discriminator value.

Discriminator weight should therefore remain separate from general relationship strength.

15. Contradictory Diagnosis Context

A Diagnosis–Evidence Relationship may identify diagnoses contradicted or weakened by the evidence.

This should support:

- differential elimination
- clue-progression analysis
- explanation generation
- diagnostic comparison
- case validation

Contradictory diagnosis references should resolve to governed registry identities where possible.

16. Supporting Artifacts

A Diagnosis–Evidence Relationship may be supported by:

- a Teaching Relationship
- a Teaching Rule
- a Clinical Case
- a Graph Fact
- external evidence

Supporting artifacts explain why the relationship exists or how it is taught.

They do not automatically establish scientific authority.

17. Diagnosis–Evidence Relationship Lifecycle

The canonical lifecycle is:

Candidate

↓

Review

↓

Active, Rejected, or Deprecated

17.1 Candidate

The relationship has been proposed but remains ungoverned.

17.2 Active

The relationship has been accepted for operational use.

17.3 Rejected

The proposed relationship has not been accepted.

17.4 Deprecated

The relationship is preserved historically but should not influence current editorial work.

18. Teaching Relationship

18.1 Purpose

A Diagnosis Teaching Relationship defines how two diagnoses should be compared, contrasted, clustered, or taught together.

It represents an educational relationship rather than a simple clinical association.

18.2 Directionality

Teaching Relationships are directional.

The source diagnosis is the diagnosis being taught.

The target diagnosis is the diagnosis used for contrast or connection.

For example:

Source: Chronic Pulmonary Aspergillosis
Target: Pulmonary Tuberculosis
Purpose: Teach discriminator

The reverse direction may require different teaching content and should not be assumed automatically.

18.3 Relationship types

Canonical Teaching Relationship types include:

- Differential Discriminator
 - Mimic Confusion
 - Shared Presentation
 - Escalation Contrast
 - Management Contrast
 - Investigation Contrast
 - Complication Relationship
-

18.4 Differential discriminator

Identifies the key findings that help distinguish two plausible diagnoses.

The relationship should state:

- why both diagnoses are considered
 - which finding favours the source diagnosis
 - which finding favours the target
 - what uncertainty remains
-

18.5 Mimic confusion

Describes a common learner or clinical confusion between diagnoses.

It should identify:

- shared presentation
 - reason for confusion
 - harmful assumption
 - corrective discriminator
-

18.6 Shared presentation

Identifies a presentation shared by the diagnoses.

A shared-presentation relationship should not imply equivalence.

Its value lies in teaching how the same presenting problem can arise from different diseases.

18.7 Escalation contrast

Compares how urgency, deterioration, or referral differs between diagnoses.

It may teach:

- which red flags demand urgent action
 - why similar symptoms carry different risks
 - what deterioration means in each diagnosis
-

18.8 Management contrast

Compares important treatment differences.

This relationship is especially valuable where confusing the diagnoses would lead to:

- inappropriate treatment
 - delayed treatment
 - contraindicated treatment
 - missed definitive intervention
-

18.9 Investigation contrast

Compares how tests are selected or interpreted across diagnoses.

It may identify:

- different first-line tests
 - characteristic findings
 - limitations
 - false reassurance
 - confirmatory strategies
-

18.10 Complication relationship

Describes a meaningful relationship between diagnosis complications.

It should clarify whether:

- both diagnoses cause similar complications
 - one diagnosis is a complication of another
 - complication patterns help distinguish them
 - delayed diagnosis produces different outcomes
-

19. Teaching Purpose

Every Teaching Relationship should possess an explicit teaching purpose.

Canonical purposes include:

- Teach Discriminator
- Prevent Common Error
- Build Differential Cluster
- Support Case Generation
- Support Education
- Support Recall

A relationship may be clinically true but unsuitable for a particular teaching purpose.

Purpose should therefore be independently reviewed.

20. Teaching Relationship Content

A Teaching Relationship may contain:

- discriminator summary
- common confusion reason
- learner pitfall
- suggested teaching-rule key
- supporting graph fact
- supporting differential link
- supporting teaching rule
- relationship strength
- governance status

Each field should contribute to how the relationship is used.

21. Teaching Relationship Lifecycle

The canonical lifecycle is:

Candidate

↓

Needs Review

↓

Active, Rejected, or Deprecated

An Active Teaching Relationship may influence learner-facing content and generation.

Activation should require human review.

22. Differential Mapping

22.1 Purpose

Differential Mapping resolves authored differential text to a governed Diagnosis Registry identity.

Mappings may originate from:

- Clinical Cases
 - Case Revisions
 - Diagnosis Education
 - Education Revisions
-

22.2 Mapping states

A mapping may be:

- Resolved
- Ambiguous
- Unresolved
- Rejected

These states describe identity resolution.

They do not determine the clinical quality of the differential itself.

22.3 Resolution standard

A mapping should consider:

- canonical name
- accepted aliases
- abbreviation
- normalised terminology
- clinical context
- subtype
- source section
- authorial intent

Automatic string similarity should not override clinically meaningful distinctions.

22.4 Ambiguous mapping

A mapping is ambiguous where more than one registry diagnosis could reasonably match the text.

Ambiguity should be surfaced for editorial review.

The system must not select a diagnosis silently where the choice would alter clinical meaning.

22.5 Unresolved mapping

A mapping is unresolved where no suitable registry entry can be identified.

Resolution may require:

- clarification of the source text
 - selection of an existing diagnosis
 - creation of a registry candidate
 - rejection of the differential
 - preservation as non-registry text where policy permits
-

23. Differential Link

23.1 Purpose

A Differential Link represents the governed connection between a source artifact and a resolved differential diagnosis.

It allows WEOS to use the relationship in:

- reasoning analysis
 - graph development
 - coverage assessment
 - case comparison
 - education governance
 - generation context
-

23.2 Differential roles

Canonical roles include:

- Primary Mimic
 - Differential
 - Important Exclusion
 - Teaching Differential
-

23.3 Primary mimic

A Primary Mimic is a diagnosis central to the reasoning challenge.

It normally requires:

- shared presentation
 - explicit discriminator
 - explanation coverage
 - appropriate clue-progression treatment
-

23.4 Important exclusion

An Important Exclusion may be less likely but clinically dangerous to miss.

It should be governed according to its safety relevance.

23.5 Teaching differential

A Teaching Differential is included primarily because it reinforces an important educational comparison.

It should still remain clinically plausible.

24. Diagnostic Comparison

24.1 Purpose

A Diagnostic Comparison is the structured educational interpretation of differences between diagnoses.

It may be assembled from:

- Teaching Relationships
 - Evidence Relationships
 - Differential Links
 - Teaching Rules
 - Graph Facts
 - Reasoning Paths
-

24.2 Comparison dimensions

A comparison may address:

- epidemiology
- onset
- time course
- symptoms
- examination
- laboratory findings
- imaging
- discriminator
- management
- escalation
- complications

Only dimensions that meaningfully distinguish the diagnoses should be included.

24.3 Comparison quality

A strong comparison should:

- begin from genuine shared uncertainty
 - identify the highest-value discriminators
 - avoid artificial binary distinctions
 - acknowledge overlap
 - avoid absolute wording where exceptions exist
 - remain applicable beyond one memorised table
-

25. Reasoning Path

25.1 Purpose

A Reasoning Path is a governed diagnosis-level specification for how a particular reasoning objective should be taught or generated.

It coordinates the knowledge required to produce a coherent case, education section, teaching rule, or recall activity.

25.2 Reasoning goals

Canonical Reasoning Goals include:

- Differential Discrimination
- Escalation Recognition
- Management Contrast
- Complication Recognition
- Early-Presentation Recognition
- Resource-Limited Reasoning

Each path should focus on one primary goal.

25.3 Generation purposes

A Reasoning Path may support:

- Case Generation
- Teaching-Rule Generation
- Education Generation

- Recall Generation

Generation purpose identifies where the path may be used.

It does not determine whether generated content is approved.

25.4 Required structure

A Reasoning Path may define:

- stable key
 - title
 - diagnosis
 - reasoning goal
 - primary differentials
 - supporting Teaching Relationships
 - supporting Evidence Relationships
 - discriminator Evidence Nodes
 - escalation Evidence Nodes
 - contradictory Evidence Nodes
 - required teaching points
 - forbidden evidence patterns
 - recommended clue distribution
 - generation purpose
 - readiness score
 - governance status
 - reviewer
-

26. Primary Differentials

Primary differentials identify the diagnoses that create the central reasoning challenge.

They should be selected because they:

- share an important presentation
- are commonly confused
- are dangerous to miss
- teach a valuable discriminator
- align with the reasoning goal

A reasoning path should not contain an unlimited differential list.

27. Supporting Relationships

Reasoning Paths should reference governed Teaching and Evidence Relationships where available.

This prevents the same reasoning from being recreated inconsistently in every generation request.

Supporting relationships should be active and relevant to the path's purpose.

28. Discriminator Evidence

Discriminator Evidence Nodes identify the findings expected to separate the diagnosis from its primary alternatives.

The path should clarify:

- which diagnosis the evidence favours
 - what it argues against
 - expected strength
 - appropriate clue timing
 - whether it is required or optional
-

29. Escalation Evidence

Escalation Evidence identifies findings that should trigger urgent action, referral, monitoring, or intervention.

It should be included where escalation recognition is part of the reasoning objective.

30. Contradictory Evidence

Contradictory Evidence identifies findings that should weaken the target diagnosis or raise concern about case coherence.

This enables generation and validation to detect when a draft accidentally undermines its intended answer.

Contradictory evidence should not be interpreted as absolute exclusion unless clinically justified.

31. Required Teaching Points

Required teaching points identify the concepts that generated or authored content must address.

They may include:

- a discriminator
- common error
- investigation limitation
- escalation threshold
- management contrast
- complication signal

Required teaching points should remain specific enough for validation.

32. Forbidden Evidence Patterns

Forbidden Evidence Patterns identify evidence use that would make content unsafe or educationally weak.

Examples include:

- revealing the diagnosis in the opening clue
- using a pathognomonic finding before meaningful reasoning
- combining mutually inconsistent findings
- relying on an unavailable specialist test in a resource-limited path
- using a normal result as absolute exclusion
- introducing treatment response that directly names the diagnosis
- contradicting the intended disease stage

Forbidden patterns should include rationale.

33. Recommended Clue Distribution

A Reasoning Path may specify the expected distribution of clue types.

For example:

History → Examination → Basic investigations → Discriminator

or:

Risk factor → Early presentation → Deterioration → Escalation evidence

Recommended distribution guides generation.

It should not force all cases into identical sequences.

34. Reasoning Path Readiness

A Reasoning Path should only become active when its required foundations are sufficiently complete.

Readiness may consider:

- resolved diagnosis
- defined reasoning goal
- resolved primary differentials
- active supporting relationships
- sufficient discriminator evidence
- required teaching points
- absence of critical contradictions
- appropriate generation purpose
- human review

A readiness score is an assessment.

It is not an approval decision.

35. Reasoning Path Lifecycle

The canonical lifecycle is:

Candidate

↓

Review

↓

Active, Rejected, or Deprecated

35.1 Candidate

The path is under development.

35.2 Active

The path may influence generation and assessment.

35.3 Rejected

The path has not been accepted.

35.4 Deprecated

The path should no longer guide new work but remains historically traceable.

36. Reasoning Draft Validation

36.1 Purpose

Reasoning Draft Validation assesses whether generated or authored artifacts correctly apply governed reasoning context.

It may evaluate:

- Cases
 - Teaching Rules
 - Diagnosis Education
 - Individual Education Sections
-

36.2 Validation outputs

Reasoning Draft Validation may produce:

- trust score
- trust tier
- validation status
- blockers
- warnings
- strengths
- hallucination-risk signals

- reasoning coverage
 - evidence coverage
 - discriminator coverage
 - unsupported-claim signals
 - recommendations
-

36.3 Trust tiers

Canonical Trust Tiers include:

- High Trust
- Review Required
- Low Trust
- Blocked

Trust tier communicates assessment confidence and risk.

It does not authorise approval or publication.

36.4 Validation statuses

Canonical Validation Statuses include:

- Passed
- Needs Review
- Failed

A passed validation means the automated assessment found no defined blocking failure.

It does not establish full clinical correctness.

37. Unsupported-Claim Signal

37.1 Purpose

An Unsupported-Claim Signal identifies a statement that may lack sufficient support from the available governed context or external evidence.

It is a quality and governance signal.

It is not automatically proof that the claim is false.

37.2 Signal sources

Unsupported-claim signals may arise from:

- contradiction with Teaching Rules
 - absence from active Evidence Relationships
 - mismatch with Reasoning Path
 - lack of supporting reference
 - overstatement of certainty
 - management claim without context
 - invented discriminator
 - ungrounded AI output
 - inconsistency with diagnosis metadata
-

37.3 Claim classification

Signals should distinguish:

- factual clinical claim
- diagnostic interpretation
- management recommendation
- prognostic statement
- scoring-system statement
- epidemiological statement
- educational generalisation

Different claim types require different forms of support.

37.4 Signal severity

Severity may consider:

- potential learner harm
- clinical consequence
- degree of contradiction
- confidence of detection
- publication status
- visibility
- repetition across artifacts

A minor unsupported educational phrase should not be treated identically to unsafe management guidance.

37.5 Resolution

An Unsupported-Claim Signal may be resolved by:

- adding supporting evidence
- revising the claim
- narrowing its scope
- adding qualification
- linking an active relationship
- rejecting the draft
- documenting an accepted editorial rationale
- removing the claim

Resolution should preserve the original signal and decision history.

38. Knowledge Contradiction

A Knowledge Contradiction exists where governed or candidate artifacts make clinically incompatible claims.

Contradiction analysis may compare:

- Case and Diagnosis Education
- Teaching Rule and Case
- Graph Fact and Reference
- Evidence Relationship and Explanation
- Management guidance across sections
- Scoring-system use across diagnoses
- active and deprecated knowledge

Contradictions should be surfaced for review.

The system should not silently choose one version as correct without governance.

39. Knowledge Coverage

Knowledge Coverage evaluates whether the diagnosis possesses sufficient governed relationships for its intended editorial use.

Coverage may assess:

- findings
- investigations

- mimics
- discriminators
- escalation signals
- management signals
- complications
- pitfalls
- reasoning paths
- recall relationships

Coverage should be purpose-specific.

A diagnosis may possess enough knowledge for education generation but not enough for advanced case generation.

40. Knowledge Readiness

Knowledge Readiness should identify the operation being evaluated.

Examples include:

- ready for graph review
- ready for reasoning-path creation
- ready for case generation
- ready for differential teaching
- ready for recall generation
- ready for publication assessment

Readiness should identify:

- available foundations
- missing relationships
- blockers
- warnings
- recommended next actions

41. Knowledge-Governance Review

Review of knowledge artifacts should assess:

- clinical accuracy
- appropriate scope
- diagnosis identity
- relationship direction

- relationship type
- teaching purpose
- source provenance
- generalisability
- contradiction risk
- evidence support
- educational usefulness
- duplicate risk
- operational consequences

Graph review should not be reduced to accepting or rejecting extracted text.

42. Governance Decisions

Knowledge Governance may produce decisions including:

- approve candidate
- reject candidate
- merge candidate
- promote candidate to fact
- archive fact
- activate evidence node
- reject evidence node
- activate evidence relationship
- deprecate evidence relationship
- activate teaching relationship
- request relationship changes
- reject teaching relationship
- activate reasoning path
- deprecate reasoning path
- resolve differential mapping
- reject differential mapping
- approve differential link
- resolve unsupported claim
- request evidence
- repair contradiction

Every decision should preserve:

- reviewer
- rationale
- affected artifact
- affected version where relevant
- source context
- resulting state
- date and time

43. Human Authority

Automated systems may:

- extract Graph Candidates
- suggest target diagnoses
- estimate confidence
- detect duplicates
- propose Evidence Relationships
- identify Teaching Relationships
- generate Reasoning Paths
- calculate readiness
- detect contradictions
- flag unsupported claims
- recommend editorial action

Automated systems must not independently:

- promote candidate knowledge into institutional fact
- activate diagnosis relationships
- resolve clinically ambiguous diagnosis mappings
- approve reasoning paths
- dismiss high-risk unsupported claims
- publish learner-facing consequences

Human reviewers retain authority over governed knowledge.

44. Knowledge Maintenance

Knowledge maintenance may be triggered by:

- new clinical evidence
- guideline changes
- diagnosis deprecation
- terminology changes
- repeated learner confusion
- case inconsistency
- education revision
- duplicate detection
- graph contradiction
- new important mimics
- changed management
- changed scoring systems
- specialty review

Maintenance should assess downstream impact.

Changing one relationship may affect:

- cases
 - education
 - recall prompts
 - reasoning paths
 - validation rules
 - publication readiness
-

45. Provenance Preservation

Every active knowledge artifact should remain traceable to its origin and governance history.

Provenance should answer:

- Where did this knowledge originate?
- Which artifact and version contained it?
- Was it extracted, generated, imported, or manually authored?
- Who reviewed it?
- Which decision activated it?
- Has it been revised, merged, or deprecated?
- Which learner-facing artifacts depend upon it?

Provenance is essential to institutional trust.

46. Quality Standard for Governed Knowledge

High-quality governed knowledge should be:

Clinically accurate

The knowledge is defensible within its stated context.

Explicitly scoped

Qualifiers, population, stage, and setting are preserved.

Diagnosis-resolved

Diagnosis relationships use governed identities.

Reasoning-centred

Relationships explain how evidence changes clinical interpretation.

Differentially useful

Important mimics and discriminators are represented meaningfully.

Provenance-preserving

The source and review history remain traceable.

Non-duplicative

Equivalent knowledge is consolidated without erasing source history.

Educationally purposeful

Relationships support a defined teaching or reasoning function.

Operationally safe

Only governed knowledge influences generation, validation, and learner-facing publication.

Maintainable

Knowledge may be revised, deprecated, merged, or archived without erasing history.

47. Normative Statements

- Extracted knowledge **MUST** begin as candidate knowledge.
- Candidate knowledge **MUST NOT** be treated as a governed fact before editorial approval.
- Every Graph Candidate **MUST** preserve its source artifact, source version, and source location.
- Candidate normalisation **MUST NOT** remove clinically meaningful qualifiers.
- Candidate merging **MUST** preserve the provenance of all contributing candidates.
- Every active Graph Fact **MUST** possess governed provenance.
- Evidence Nodes **MUST** remain distinct from Diagnosis–Evidence Relationships.
- An active Evidence Node **MUST NOT** imply an active relationship with any diagnosis.
- Diagnosis–Evidence Relationships **MUST** state how the evidence affects reasoning.
- `RULES_OUT` relationships **MUST** be used conservatively and clinically defensibly.
- Relationship strength **MUST NOT** be treated as equivalent to discriminator value.
- Teaching Relationships **MUST** preserve source and target diagnosis direction.

- Every Teaching Relationship **MUST** possess an explicit teaching purpose.
 - Active Teaching Relationships **MUST** undergo human review.
 - Differential mappings **MUST NOT** silently resolve clinically ambiguous text.
 - Primary mimics **SHOULD** possess explicit discriminator coverage.
 - Reasoning Paths **MUST** define a primary reasoning goal.
 - Reasoning Paths used for generation **MUST** identify their generation purpose.
 - Forbidden Evidence Patterns **MUST** include clinically or educationally meaningful rationale.
 - Reasoning readiness scores **MUST NOT** be interpreted as approval.
 - Automated validation **MUST NOT** replace clinical editorial review.
 - Unsupported-Claim Signals **MUST** remain distinguishable from confirmed factual errors.
 - High-risk unsupported management claims **MUST** block publication until resolved.
 - Knowledge contradictions **MUST** be surfaced rather than silently overwritten.
 - Only governed knowledge **SHOULD** influence production generation context.
 - Deprecated and archived knowledge **MUST** remain historically traceable.
 - Knowledge-governance decisions **MUST** preserve reviewer accountability and rationale.
 - AI-generated relationships and reasoning paths **MUST NOT** become active without human governance.
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Closing Principle

Clinical reasoning cannot be governed if it remains implicit.

Wardle must be able to identify which findings support a diagnosis, which findings distinguish it from its mimics, which signals change urgency, which relationships shape learner understanding, and where every claim originated.

Knowledge and Reasoning Governance transforms isolated educational content into a connected, explainable, and maintainable clinical knowledge system.

Its objective is not merely to connect diagnoses and findings.

Its objective is to ensure that every connection used to teach, generate, assess, or publish diagnostic knowledge is clinically meaningful, educationally purposeful, and institutionally governed.